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Managing Information Security

A1. Stellar Financial Solutions is a financial institution that is a private company with its headquarters in San Francisco and offices within the European Union. The two types of cybersecurity incidents that need to be addressed within the incident response plan are the ransomware and phishing attacks, as these have both been experienced by the company. The phishing attack that had occurred compromised the information of the company's employees and customers. The attack was done by impersonating the HR department and installing Trojans that would access customer data for two months. In the ransomware attack, the ALPHV/BlackCat group used an exploit from a missing Windows patch that the company had not resolved, which allowed them to deploy ransomware that left the company incapable of operating its credit services for more than two weeks.

Stellar Financial Solutions can detect phishing attacks better by implementing an email security gateway that has anti-phishing features. The tool will be able to scan all inbound emails for any unusual signs such as spoofed sender domains, phishing sites, malicious links, and it would do so right before it reaches any employee's inbox. These alerts can also be reported by the SIEM so that these cases can be properly investigated in a timely manner without going unnoticed. As for the ransomware, Stellar Financial Solutions needs to begin leveraging its endpoint detection and response (EDR) tool across all of the company's systems. This helps the company perform an analysis that can identify any early signs of a ransomware attack such as DLL injections and callbacks to malicious domains that may otherwise go undetected.

To contain incidents such as the phishing attack that compromised credentials of users, it should begin by disabling any affected accounts and forcing users to reset their passwords. Then it should begin to separate the users from the network. This will prevent attackers from continuing to use stolen credentials and from moving laterally throughout the network. Another way that they could begin containing the ransomware incident would be by isolating the affected network segments by using the EDR tool I mentioned previously to clean up any of the infected hosts.

This will help limit the attacks by preventing infection points and preventing the spread of attacker encryption across servers.

These containment recommendations align with Stellar Financial Solutions' security goals identified in the case study. Disabling compromised accounts and separating the affected users from the network by segmenting them from the rest of the users supports the company's goal of establishing an effective incident response plan that incorporates business continuity and disaster recovery planning, since cutting off the attackers' ability to gain access early stops a single compromised account from turning into a larger breach that disturbs operations within the company. Isolating the affected network through segmentation during a ransomware attack supports the company's goal of improving its overall system availability across the network because it prevents the spread of the attack and keeps it contained. This also allows the company to mitigate any damage and begin to create a recovery and incident plan.

A2. The first recommendation I would like to make for Stellar Financial Solutions is an incident response manager being implemented as a role within the company who would be responsible for maintaining incident response plans, communicating with leadership about incidents, doing incident reviews after they happen, and making sure the team is taking action when these incidents happen. This role will help promote a more positive response to any incidents that happen. Based on the case study Stellar Financial Solutions is currently only conducting incident response activities with the network administrators and the office's IT staff. There is no dedicated role to manage incident responses.

The second recommendation I would like to suggest is that Stellar Financial Solutions should add a security operations center (SOC) analyst role within the company that will be responsible for detecting incidents by using tools such as SIEM and EDR around the clock, determining which threats are real or false positives, and escalating any incidents that happen to the incident response manager. Based on the case study the company currently only monitors alerts and threats with the IT staff which is completely secondary to their main work duties. This is why the phishing attack went undetected for two months, as the staff were busy with other work and didn't give it as much attention as a SOC analyst would.

These roles at Stellar Financial Solutions would help the company maintain business operations with strong roles dedicated to specific tasks. Currently they are mixing different duties with the wrong roles and it's causing the company to be vulnerable to certain attacks because it isn't dedicating tasks to specific roles around the clock. By making this implementation it will give each role a dedicated position for specific tasks and it takes away the work load from other positions whose responsibility should be focused on other things. The incident response manager helps the company maintain business operations by making sure all incidents are handled with a dedicated team that is making sure things are organized. The SOC analyst will conduct consistent monitoring of any alerts that come through the SIEM and EDR, which will detect attacks like phishing in their early stages before they can disrupt the company's operations. These implementations will keep the company on top of things.

A3. A metric that would help Stellar Financial Solutions would be tracking the amount of time the incident response team takes to respond to an incident from the time it's detected to the time when it's officially contained (Cichonski et al., 2012). This will ensure that the company can consistently improve its procedures by seeing over time how long it takes to effectively contain incidents.

A4. The incident response plan created by Stellar Financial Solutions needs to be communicated with stakeholders by explaining how it supports the company's security goals. All positions of leadership should receive a summary of the plan, including the costs and how it supports the company's overall goals when it comes to security and minimization of risk. The incident response team and IT staff should also receive detailed training that includes exercises so they can continue to be competent when it comes to understanding their responsibilities around incidents. All other employees should also be educated about the plan and should be aware of any possible security breach instances and how to report them when they suspect something unusual. This plan needs to be kept in an accessible place where all staff can access it, and stakeholders will be updated when any changes occur.

References

Cichonski, P., Millar, T., Grance, T., & Scarfone, K. (2012). *Computer security incident handling guide* (NIST Special Publication 800-61 Rev. 2). National Institute of Standards and Technology.
<https://doi.org/10.6028/NIST.SP.800-61r2>